

22k-4785 Syed Shaiq Hussain

22k-4782 Mavia Mehmood

EHTT Project Report: Advanced Keylogger and Surveillance Tool

This project is an advanced monitoring and control suite designed for system surveillance, data collection, and remote command execution. This project integrates multiple functionalities, ranging from keystroke logging and clipboard monitoring to video recording and WiFi credential retrieval. It can be utilized for advanced diagnostics, user activity monitoring, and remote administrative controls.

Features

1. Keystroke Logging

- Captures and logs all keystrokes pressed on the system.
- Stores the logs in a timestamped file for review.
- Provides valuable insights into user behavior and input activity.

2. Clipboard Monitoring

- Tracks and logs clipboard activity, capturing any copied content.
- Useful for recording sensitive or temporary data.

3. Screen Recording

- Continuously records the system screen.
- High-quality video output in .avi format for playback and analysis.
- Adjustable frame rate for optimized performance.

4. Front Camera Recording

- Activates and records video using the system's front camera.
- Automatically selects the available camera hardware for seamless recording.

5. WiFi Credential Retrieval

- Retrieves saved WiFi credentials from the system.
- Compiles a list of SSIDs and associated passwords.
- Saves the data securely for reference.

6. System Information Logging

- Gathers and logs critical system information:
 - Hostname
 - Private IP address
 - Public IP address
- Stores the details in a structured file format.

7. Remote Command Execution

- Fetches and executes commands from a Google Drive-hosted file.
- Supported commands:
 - **stop**: Terminates all operations gracefully.
 - **start_front_camera**: Activates front camera recording.
 - **lock_screen**: Locks the system screen.

8. File Upload and Sharing

- Uploads log files and recordings to **File.io** for easy access and sharing.
- This method was chosen primarily because **Gmail** has a 25MB attachment size limit, and larger files would be rejected. File.io offers a seamless and efficient method for handling large files, ensuring they are stored and accessible without email size constraints.

9. Email Notifications

- Sends collected data, including keylogs, WiFi credentials, and system information, via email.
- Includes downloadable links for video recordings and other files.

10. **Security and Reliability**

- Error handling ensures smooth operation in case of unexpected issues.
- Logs all actions, errors, and activities for debugging and monitoring.

Technical Components

1. Languages and Libraries

- Python programming language.
- Libraries used:
 - os, logging, time, threading: Core functionality and threading.
 - pyperclip, pyautogui, cv2: Clipboard, screen capture, and video recording.
 - requests: Fetching remote files and APIs.
 - ctypes, subprocess: System operations like locking the screen and retrieving WiFi credentials.
 - smtplib, email: Email notifications.
 - pynput: Keystroke logging.

2. Cloud Integration

- Uses **Google Drive** for remote command hosting.
- Utilizes **File.io** for secure file uploads.

3. Output Storage

- Log files stored locally in the logs directory:
 - keylog.txt
 - clipboard.txt
 - wifi_credentials.txt
 - system_info.txt
- Video files stored locally in .avi format:
 - screen_recording.avi
 - front_camera_recording.avi

Potential Applications

1. Administrative Controls

- Monitor user activities in corporate or educational environments.
- Gather data for security audits and diagnostics.

2. Parental Monitoring

- Oversee children's online activities to ensure safe usage.

3. System Troubleshooting

- Record screen activity to debug errors or replicate user-reported issues.

4. Remote Surveillance

- Command-based control for real-time monitoring and lock-screen functionality.

5. Other Purposes.

Challenges and Solutions

1. Camera Compatibility

- The program checks multiple indices to identify available cameras, ensuring reliable video capture.

2. File Upload Reliability

- Integrated error handling for File.io uploads to prevent loss of data.

3. Security Concerns

- Ensures data is stored locally and shared securely using encrypted email and File.io links.